Mekong Quality Assurance Advisory

Keeping production true to what you approved

How the Product Specification, Packaging Specification and Verification Protocol fit together, and who owns each.

You approved a product. The risk in co-packing is that later production runs slowly stop matching it: a little darker, a little thinner, a different cap, a label with a printing error. Nobody changed the recipe on purpose, and nobody can prove what the standard was, because it was never written down tightly enough to check. I call that drift.

I prevent it with three working documents. The **Product Specification** fixes what your approved product is, as measurable ranges. The **Packaging Specification** fixes what the pack, label and pallet must be. The **Verification Protocol** is how I check each production run against both. Your counsel attaches them to the agreement with your co-packer, so they are enforceable.

I work from your approved finished product, not from your recipe. The recipe stays yours.

Naming. Your own paperwork may call these documents a schedule, annex or exhibit. What matters is that the content exists.

Scope. Results are reported against your specification. This is not a food safety certification. The plant controls food safety; I provide independent verification.

The four documents at a glance

Document	What it is	Drift it prevents
Master Quality Agreement	The signed agreement between you and the co-packer that the other three documents attach to, written by your counsel: consequences of failure, change control, per-lot certificate of analysis, retained samples, traceability, recalls.	Nothing makes the plant answerable when a lot fails, and changes happen without sign-off.
Product Specification	The food: color, °Brix, pH, water activity, consistency, fill, visual, and taste and smell, as measurable ranges built from your approved product.	The product slowly moving away from the one you approved.
Packaging Specification	The pack: seal and closure, label and printing, ingredient list and allergen statement as printed, barcode, lot and date code, carton, pallet.	Quiet changes in materials and pallets, and printing errors that reach the shelf.
Verification Protocol	How each production run is checked against the other two: what I measure on site, what goes to a lab, sampling, defect classes, what happens when a lot fails.	The first two documents sitting on paper with nobody checking them.

The big picture

Teal: I write it or carry it out. Amber: you own or approve it. Gray: the plant or your regulatory advisor owns it.

MASTER QUALITY AGREEMENT

Owned by you and your counsel

Makes the three documents below binding. Holds consequences of failure (reject, retest, hold), change control, per-lot CoA, retained samples, traceability, recalls.

Drift prevented: nothing makes the plant answerable when a lot fails.

PRODUCT SPECIFICATION

I write it; you approve

The food: quality and consistency (safety limits are not set here)

- Color range (L*a*b*), plus physical color tiles
- °Brix, pH and water activity ranges (consistency indicators)
- Bostwick consistency, fill, visual (specks, gloss, separation)
- Taste and smell against a retained reference sample
- Named off-notes, zero tolerance: burnt, metallic, rancid, fermented
- One standard test temperature; target plus range
- Initial ranges tighten by an agreed rule after the first lots

Built from your approved product, not your recipe.

Drift prevented: the product slowly moving away from the original.

PACKAGING SPECIFICATION

You own the standard; I record it

The pack: bottle, closure, label, carton, pallet

- Path 1: your packaging consultant writes it
- Path 2: the plant states its specs, I record them, you approve
- Pack and seal, cap torque, cartons, pallet pattern and wrap
- Print quality and printing errors; barcode; lot and date code
- Ingredient list and allergen statement as printed, matched to your approved artwork (legal compliance is for your advisor)
- Light document check; not a structural engineering review

Drift prevented: quiet changes in materials, print and pallets.

VERIFICATION PROTOCOL

I write it and carry it out

ON SITE, with my own instruments

- Bostwick, refractometer (°Brix), pH meter, water activity meter
- Color: plant's spectrophotometer after a calibration check; otherwise tiles in a D65 light booth (lower confidence)
- Taste and smell vs. reference; blind panel for quality calls
- Fill, seals, labels, print, codes; cartons, pallets, foreign matter
- Samples brought to one standard temperature before testing
- AQL sampling (ANSI/ASQ Z1.4); critical, major, minor defects

SENT TO AN ACCREDITED LAB (about 3 to 4 days)

- Color baseline readings when the specification is set
- Microbiology and other lab-only tests you ask for
- Disputed readings: retained split sample to a pre-agreed lab
- I collect, seal, ship and track every sample; the lab tests; results are reported against your specification
- Lot held until lab results are back
- One tamper-sealed sample kept from each inspection

Inspection report to you, with readings and photos

You decide: release, hold, rework or reject

FOOD SAFETY: owned by the plant and your regulatory advisor

HACCP, process validation, certification, label legality. I don't validate or certify; I report and escalate anything safety-related I see.

Who owns what

Component	Prepared by	Approved by	My role	Not my job
Recipe and formulation	You	You	I work from the finished product you approve, not from the recipe.	Writing, changing or reviewing your recipe.
Master Quality Agreement	Your counsel	You and the co-packer	I supply the three documents to attach, and advise on what the agreement should require.	Drafting legal terms or giving legal advice.
Product Specification	Me	You (the co-packer acknowledges it)	I set the ranges from baseline readings, make the color tiles, and set the taste and smell reference.	Setting food safety limits.
Packaging Specification	Path 1: your packaging consultant. Path 2: the co-packer states its specs.	You	Path 1: I turn it into inspection checks. Path 2: I request the plant's specs, record them, and run a light document check.	Engineering or designing the packaging, or certifying it will survive transit.
Verification Protocol	Me	You	I write it, carry out the inspections, and report the results.	Releasing or rejecting the lot. You decide.
Food safety and label legality	The co-packer; your regulatory advisor	Certification body or regulator, where applicable	I report and escalate any safety-related finding.	Validating or certifying food safety, or ruling on whether a label is compliant.

Product Specification: what goes in it

The baseline comes from your approved product, read with my own instruments and, for color, an accredited lab. Each parameter has a target value and a range. Every test is run at one standard temperature, 20 °C or 25 °C, written into the specification. Hot samples are brought to temperature in a water bath first.

Initial ranges are time-boxed. For the first commercial lots (for example, the first three, or the first run if you produce less often), the ranges are initial ranges: they are outer limits and never loosen. After those lots they tighten by a method agreed in advance, such as recalculating from the first lots' data with a maximum width, and the tightened ranges take effect automatically, with no renegotiation. The target value stays fixed, and my reports track where each lot lands inside its range, so a plant drifting toward the loose edge is flagged even while lots still pass. Your counsel writes the automatic-tightening clause into the agreement.

Parameter	How the range is set	How I check it at inspection	Notes
Color (L*a*b* and tiles)	Spectrophotometer readings from an accredited lab; ranges set from the results; physical tiles made: a factory set, your set, and my set.	The plant's spectrophotometer, but only after a calibration check in front of me: the plant's white tile, then a read of my reference tile, which must match its recorded values within a stated tolerance. If the check fails, or there is no instrument, I use visual comparison with the tiles in a D65 light booth.	Visual-only is lower confidence, and my report says so. Tiles get a re-check interval, set once the tile material is chosen.
°Brix	Baseline readings.	Refractometer.	
pH	Baseline readings.	pH meter with temperature compensation.	A consistency indicator. The plant's own safety limit is recorded, not set, by me.
Water activity	Baseline readings.	Water activity meter.	Same treatment as pH.

Parameter	How the range is set	How I check it at inspection	Notes
Consistency	Baseline readings.	Bostwick consistometer.	Measured at the standard test temperature.
Fill and visual	Your declared fill and the approved product.	Fill check; visual check for specks, gloss, separation.	
Taste and smell	A retained sealed reference sample, plus named off-notes with zero tolerance: burnt or scorched, metallic, rancid, fermented, bitter.	Side by side with the reference. Spoilage and contamination off-notes are reported and escalated immediately. Quality calls (burnt or scorched, metallic, bitter, off the reference) go to a blind panel that includes the plant's QA person.	I smell first, and I never taste a sample that is swollen, leaking, or smells fermented or chemical. If the panel can't agree, you decide.

What stays out of the Product Specification

Item	Where it lives	Owner
Allergens and cross-contact	The plant's food safety system. The allergen statement as printed is checked against your approved artwork through the Packaging Specification.	Co-packer; you for the label
Contaminants (heavy metals, mycotoxins, pesticide residues)	Your regulatory requirements. I can arrange lab testing if you ask.	Your regulatory advisor
Microbiology	The plant controls it. The Verification Protocol has a lab lane and a lot hold.	Co-packer (control); me (sampling and reporting)
Additives, colorants, destination-market labeling rules	Your regulatory compliance.	You
Traceability and lot coding	Lot code format in the Packaging Specification; traceability duties in the agreement.	You and the co-packer
Per-lot certificate of analysis, retained samples, change control, recalls, consequences	Master Quality Agreement.	Your counsel

Packaging Specification: two ways to build it

	Path 1: you have a packaging consultant	Path 2: you use the co-packer's packaging
Who defines the standard	Your consultant writes the Packaging Specification.	The plant states what it can do; you agree to it.
What I do	I turn the specification into inspection checks.	I request the plant's packaging specs (seal, cap torque, carton strength, pallet pattern, artwork, barcode), record them as the Packaging Specification, and run a light document check, not an engineering review: food-contact materials, whether supplier test data exists for the carton and pallet, and whether the shipping route calls for a packaging consultant or a transit test.

	Path 1: you have a packaging consultant	Path 2: you use the co-packer's packaging
Your role	You approve the consultant's specification.	You approve the recorded specification before it becomes the standard.
What gets inspected	The consultant's standard.	What the plant says it can do and you agreed to.
What the record is	The consultant's engineering, which I inspect against.	A baseline description of what the plant uses. It is not a structural load assessment, and who carries the risk of packaging that fails in transit is decided in the agreement.
What I do not do	Engineer or redesign the packaging.	Engineer or redesign the packaging.

Contents: primary pack and seal; closure and cap torque; label and artwork, including print quality, smearing, misprints and color against the approved artwork; ingredient list and allergen statement as printed; net contents and claims as approved; barcode; lot and date code; master carton; pallet pattern and wrap.

I check that what is printed matches the artwork you approved. Whether that label is legally compliant is for your regulatory advisor.

In Path 2 the record describes what the plant uses. For a long or rough shipping route, I recommend a packaging consultant or a transit test, and your counsel should decide in the agreement who carries the risk if packaging fails in transit.

Verification Protocol: what I check and how

Check	Where	Instrument or method	Notes
Consistency	On site	Bostwick consistometer	At the standard test temperature.
°Brix	On site	Refractometer	
pH	On site	pH meter with temperature compensation	Fresh calibration buffers each time.
Water activity	On site	Water activity meter	Calibration standards each time.
Color	On site	The plant's spectrophotometer, accepted only after a calibration check; otherwise tiles in a D65 light booth	Calibration check: the plant's white tile, then my reference tile read against its recorded values. Visual-only is lower confidence.
Taste and smell	On site	Against the retained reference; named off-notes. Quality calls by blind panel.	Spoilage and contamination off-notes are escalated, not put to a panel.
Fill, seals, leaks	On site	Fill check; visual and handling checks	
Labels, printing, barcode, lot and date codes	On site	Compared with the approved artwork in the Packaging Specification; ingredient list and allergen statement read against it; barcode scanned	Printing errors are recorded with photos.
Cartons and pallets	On site	Against the Packaging Specification	
Visible foreign matter	On site	Visual	
Sample temperature	On site	Thermometer	Samples are brought to the standard temperature before testing.
Microbiology and other lab-only tests	Accredited lab	Only where you ask; results take about 3 to 4 days	Lot held until results are back.

My instrument calibration records are kept and available to you, so my readings are defensible. I also check any instrument I don't own before I accept its readings.

If a reading is disputed

The first step is to measure the same sample side by side with the plant, at the same temperature, and compare calibration records. If it is still disputed, a retained split sample goes to a pre-agreed accredited lab. The binding terms, and who pays for the lab, belong in the agreement your counsel writes.

Retained samples. The plant keeps sealed retains of each lot, and the number and the time they are kept are set in the agreement, scaled to the pack size. At each inspection I also take one tamper-sealed sample, log it and hold it, so a dispute sample never rests only on the plant's own storage.

How a taste or smell call is made

Spoilage and contamination off-notes (rancid, fermented, moldy, chemical or machine oil) are not put to a vote. I report and escalate immediately, hold the lot, and send a sample for lab analysis.

Quality calls (burnt or scorched, metallic, bitter, clearly off the reference) are made by a blind side-by-side comparison with the retained reference, by at least two tasters including the plant's QA person. A triangle test, where tasters pick the odd sample out of three, shows whether the production sample is reliably different from the reference. A quality hold never rests on my palate alone. If the panel can't agree, I give you the findings and the retained sample, and you decide.

Defect classes

Class	Examples	What happens
Critical	Visible foreign matter; leaks, seal failures or swollen packs; wrong product or wrong pack; missing or wrong allergen statement; spoilage or contamination off-notes (rancid, fermented, moldy, chemical or machine oil); any reading outside the plant's own validated safety limit.	I stop and report to you immediately. Lot on hold.
Major	°Brix, pH, water activity, color or consistency out of range; quality off-notes confirmed by the blind panel (burnt or scorched, metallic, bitter, clearly off the reference); fill out of range; printing errors or wrong claims on the label; wrong or unscannable barcode; unreadable lot or date code; crushed cartons; unstable pallet.	Counts against the AQL. A failed lot is held or rejected at your decision.
Minor	Slight label tilt; small cosmetic print flaws; smudged but legible date code; scuffed master cartons.	Counts against the AQL. Noted in the report.

Sampling uses AQL under ANSI/ASQ Z1.4, with limits set by lot size and product risk.

The food safety boundary

Item	Owner	My role
Hazard analysis and process controls (HACCP)	Co-packer	None. I do not validate it.
Process validation, including thermal process	Co-packer and its process authority	None.
Safety limits, such as a maximum pH or water activity	The plant's validated process, or your regulatory advisor	I record them as supplied, compare readings to them, and report and escalate anything outside.
Microbiological control	Co-packer	I collect and send samples to a lab if you ask, and report results against your specification.

Item	Owner	My role
Destination-market regulatory compliance and label legality	Your regulatory advisor	None.

Results are reported against your specification. This is not a food safety certification.

How an engagement runs

1. **Baseline.** You provide your approved product. I take baseline readings with my own instruments and, where needed, an accredited lab, set the ranges, make the color tiles, and retain the taste and smell reference. This produces the Product Specification.
2. **Packaging.** Path 1 (your consultant writes it) or Path 2 (the plant states its specs and I record them). You approve. This produces the Packaging Specification.
3. **Protocol.** I write the Verification Protocol from the first two documents: instruments, on-site and lab lanes, AQL sampling, defect classes.
4. **Agreement.** You and your counsel attach the three documents to the Master Quality Agreement.
5. **Inspect.** I check each production run, scheduled or unannounced, and report with readings and photos. You decide to release, hold, rework or reject.
6. **Review drift.** After the first lots the ranges tighten by the method agreed in advance. I also re-check or replace the color tiles and update the documents through change control.